

Odd Mid Semester Examination 2025

Name of the Course: **Bachelor of Technology (Biotechnology)**
 Name of the Paper: **Circular Economy and Sustainability**
 Time: 1:30 Hours

Semester: **V**
 Paper Code: **TBT509c**
 Maximum Marks: 50

Note: -

- (i) All questions are compulsory.
- (ii) Answer any one question among a & b in each main question
- (iii) Total marks in each main question are ten.
- (iv) Each question carries 10 marks

Q1	(10marks)	
(a)	Fill in the Blanks 1. The traditional "take-make-waste" model is also known as the _____ economy. 2. Servitization is a business model that focuses on selling a _____ rather than a physical product. 3. One of the main limitations of the linear economy is its reliance on finite _____. 4. In the circular economy, an item like a computer motherboard that cannot be composted is referred to as a _____ nutrient. 5. A key goal of the circular economy is to eliminate the concept of _____ from the system.	CO1
	OR	
(b)	What is the difference between a biological nutrient and a technical nutrient in the context of the circular economy?	
Q2	(10 marks)	
(a)	What are the three core principles of the circular economy as defined by the Ellen MacArthur Foundation?	CO2
	OR	
(b)	How does a product service system (PSS) fundamentally change the relationship between a company and its customer?	
Q3	(10 marks)	
(a)	Analyze how the concept of planned obsolescence directly contradicts the core principles of a circular economy.	CO2
	OR	
(b)	Explain any two in short: i) extended producer responsibility (EPR) ii) remanufacturing iii) product-service system (PSS)	
Q4	(10 marks)	
(a)	What is the purpose of a closed-loop supply chain ?	CO3
	OR	
(b)	Describe what is meant by "keeping products and materials in use." Provide a specific example.	
Q5	(10 marks)	
(a)	Highlight the differences between 3R and 7R Principle? With proper reasoning explain which one is more suitable for supporting circular economy?	CO1
	OR	
(b)	Explain the relationship between the circular economy and overall sustainability. How do they support each other?	

